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M168_Notes_8.1

Chapter 8 Probability Distributions and Statistics

Section 8.1 Distributions of Random Variables

The value of the result of the probability experiment is a RANDOM VARIABLE.

If you roll a die we can let X be the number of dots showing,

$X = \{1, 2, \ldots, 6\}$.

If we have a hand of three cards, X could be the number of clubs in the hand,

$X = 0, 1, 2, 3$

These examples are all finite discrete random variables.

<u>*Example*</u>

Roll a single die and count the number of rolls until a 6 comes up.

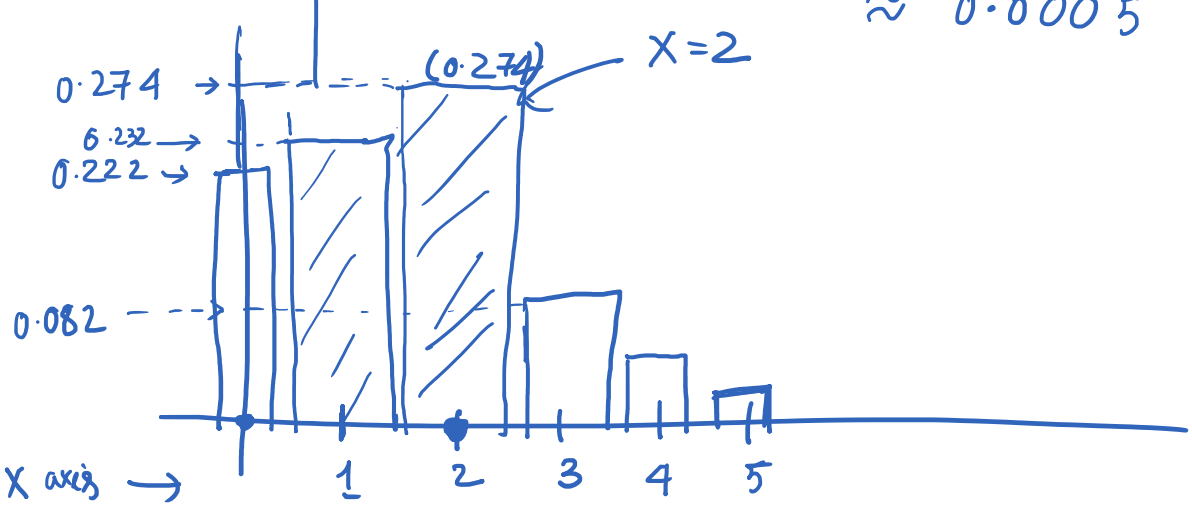
outcome	Y
6	1
1, 6	2
1, 4, 6	3
1, 2, 3, ..., 3, 6	23

X

Finally, the random variable can be continuous:

The time, in hours, spent studying for a test. $t \geq 0$

Example

You are dealt a hand of 5 cards. Find the probability distribution table for the number of hearts. Graph this in a histogram

# of Hearts x	$P(x)$
0	$\frac{C(13,0) \times C(39,5)}{C(52,5)} \approx 0.222$
1	$\frac{C(13,1) \times C(39,4)}{C(52,5)} \approx 0.232$
2	≈ 0.274
3	≈ 0.082
4	≈ 0.011
5	≈ 0.0005